ISYE 7406 Project Proposal: Airline Customer Satisfaction

Group Number : 193
Rajeev Gundavarapu(rgundavarapu3@gatech.edu) , Neha Hatolkar(nhatolkar3@gatech.edu), Avni Pherwani(apherwani6@gatech.edu)

Project Description:

The airline industry has the potential to be one of the most lucrative industries , and at the same time also comes with high capital requirements. To ensure maximum profits, it is imperative to cater to the number one aspect which can guarantee success, which is none other than customer satisfaction. For any business to be profitable, it must ensure that its customers are satisfied to ensure growth. Major Airlines in the world such as Emirates, Etihad, Qatar Airways, Delta Airlines, and American Airlines are some of the most successful airlines that the world has seen till date and they are still continuing to grow their customer base exponentially. This is because they pay immense detail to customer satisfaction. If customers are satisfied with their flying experience, they are likely to fly with the same airline next time, and can even go to the extent of advertising for it. Airlines also have to be mindful of the budget in enhancing various features of the flying experience such as booking, facilities such as wifi in the aircraft, food, etc and it is important to analyse where to invest money to ensure maximum returns. In this project, we aim to analyse different factors which are essential to contributing to a satisfying flying experience for customers with a variety of different demographics. We will analyse a variety of features that are potential factors in resulting in satisfied customers and draw conclusions as to which aspects are the most important for certain demographics. We will also provide the ability to be able to predict whether a customer will have a satisfying experience given a set of ratings for various factors that are considered as features within our dataset.

Dataset Information:

The dataset for this project was obtained from Kaggle and the url to dataset is https://www.kaggle.com/datasets/teejmahal20/airline-passenger-satisfaction . The data set is of size 15.23 MB and consists of a training and test dataset. Following are the data columns available in the data set with their description:

Attribute name	Description	Datatype	Units
Gender	Gender of the passengers	Categorical	(Female, Male)
Customer Type	The customer type	Categorical	(Loyal customer, disloyal customer)
Age	The actual age of the passengers	Numerical	Years
Type of Travel	Purpose of the flight of the passengers	Categorical	(Personal Travel, Business Travel)
Class	Travel class in the plane of the passengers	Categorical	(Business, Eco, Eco Plus)
Flight distance	The flight distance of this journey	Numerical	Miles
Inflight wifi service	Satisfaction level of the inflight wifi service	Numerical	Rating (0: Not Applicable;1-5)
Departure/Arrival time convenient	Satisfaction level of departure/arrival time to the passengers	Numerical	Rating (1-5)
Ease of Online booking	Satisfaction level of online booking	Numerical	Rating (1-5)
Gate location	Satisfaction level of Gate location	Numerical	Rating (1-5)
Food and drink	Satisfaction level of Food and drink	Numerical	Rating (1-5)
Online boarding	Satisfaction level of online boarding	Numerical	Rating (1-5)
Seat comfort	Satisfaction level of Seat comfort	Numerical	Rating (1-5)

Inflight entertainment	Satisfaction level of inflight entertainment	Numerical	Rating (1-5)
On-board service	Satisfaction level of On-board service	Numerical	Rating (1-5)
Leg room service	Satisfaction level of Leg room service	Numerical	Rating (1-5)
Baggage handling	Satisfaction level of baggage handling	Numerical	Rating (1-5)
Check-in service	Satisfaction level of Check-in service	Numerical	Rating (1-5)
Inflight service	Satisfaction level of inflight service	Numerical	Rating (1-5)
Cleanliness	Satisfaction level of Cleanliness	Numerical	Rating (1-5)
Departure Delay	Minutes delayed when departure	Numerical	Minutes
Arrival Delay	Minutes delayed when Arrival	Numerical	Minutes
Satisfaction	Airline satisfaction level	Categorical	(Satisfaction, neutral or dissatisfaction)

All of the above factors are going to be considered in measuring the satisfaction level of the customer. The fields for ratings will have values of 1-5 and features such as Departure Delay, Arrival Delay, and Flight distance are measured in minutes and kilometres respectively. Other potential factors that could potentially affect the outcome of customer satisfaction can also be added to the dataset in the future if relevant.

Scientific Research and Methodology :

Exploratory Data Analysis:

We will perform basic data analysis to understand which features are the most important. We will try to understand the relationships between different attributes and how they are correlated with each other. One potential method is to derive a correlation matrix in the form of a heatmap and also generate scatterplots to understand the relationships between selected attributes. Data are summarised using correlation matrices, which can be utilised as inputs for more sophisticated studies. One example of such can be to segment the datasets in age groups and gender and identify the most relevant features that are important to the customer. Other forms of exploratory data analysis include finding out which customer demographic tends to be the most dominant , so that budgeting can be catered to some of the most dominant groups. We will also perform basic outlier analysis to remove outliers that can potentially affect our classification.

Feature selection:

LASSO, Decision tree, information gain will be used to identify features along with our analysis of the correlation matrix. We will also explore backward feature selection and forward feature selection as well with this dataset. In case most of the features are of importance in our analysis, we will also try using dimensionality reduction techniques such as PCA for our training to capture the complete variance of the data.

Customer Segmentation and Grouping:

We will explore algorithms such as KMeans and mini batch KMeans clustering to group relevant segments of customers together for various features of customer satisfaction or components of the result of dimensionality reduction to find similarities between various demographics.

Prediction of Satisfaction Level:

We are going to explore various techniques such as classification analysis, KNN, various ensemble methods such as XGBoost, SVM, Random Forest,logistic regression to classify and find probabilities of customer satisfaction. We will choose the model that performed the best on our training data , for which we will use k-fold cross validation to train. The model which works best will be used on the test dataset, and we will be able to predict the probability classify the test data point to a customer satisfaction level.